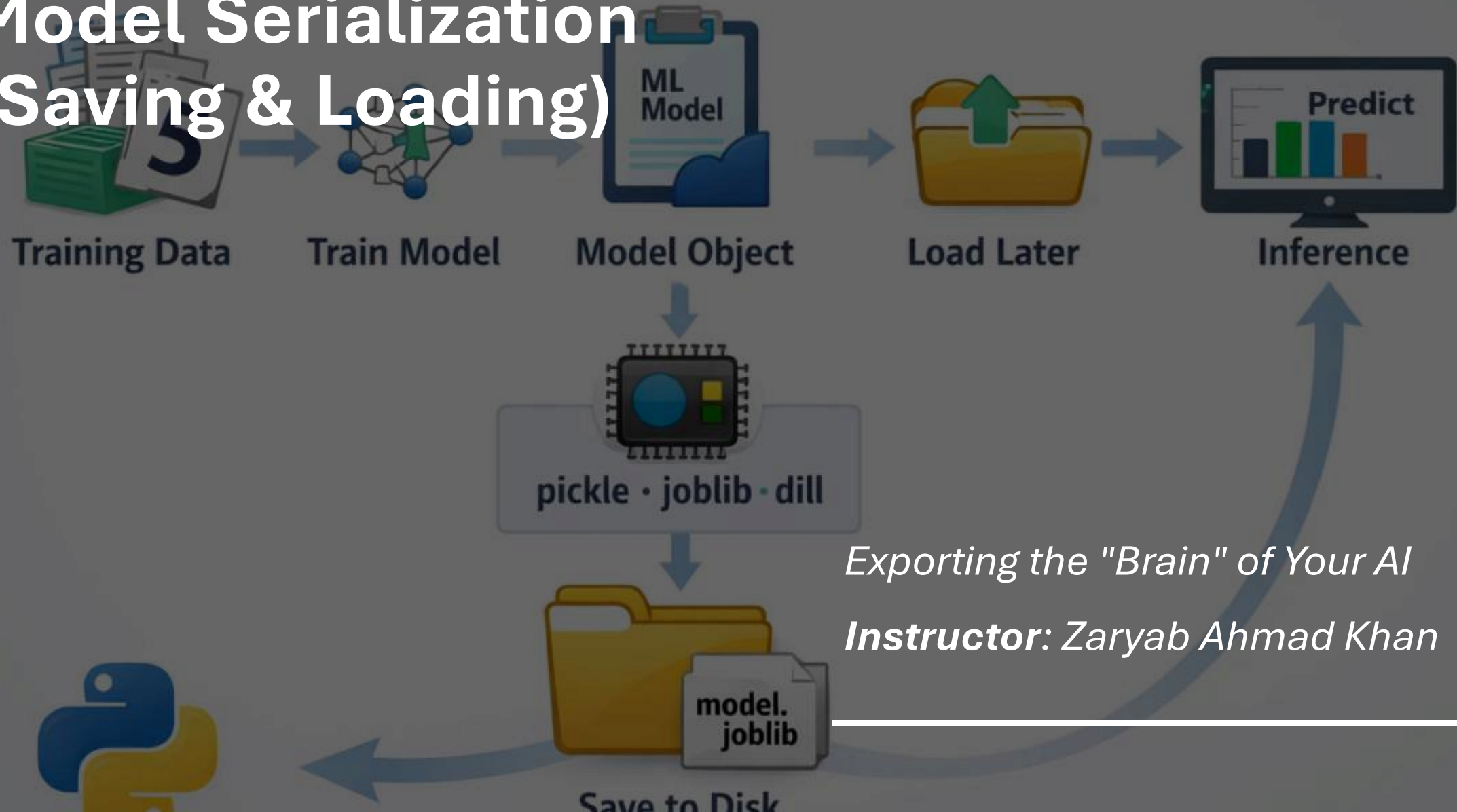
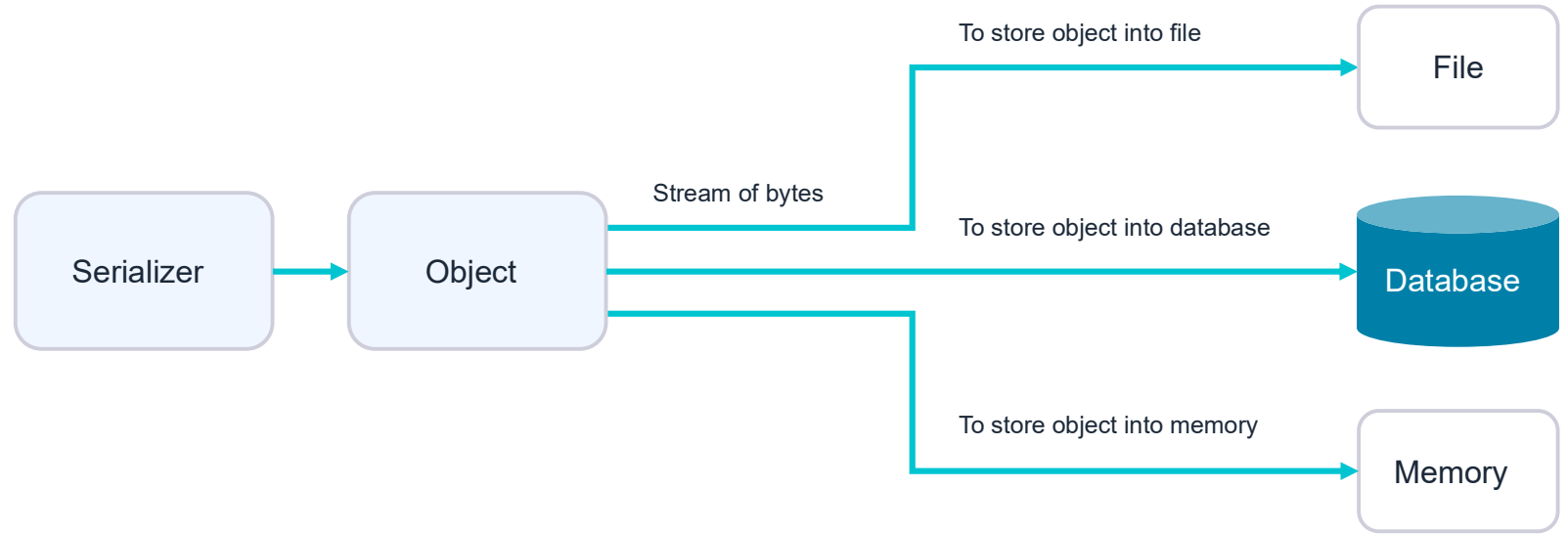


Model Serialization Workflow

Model Serialization (Saving & Loading)



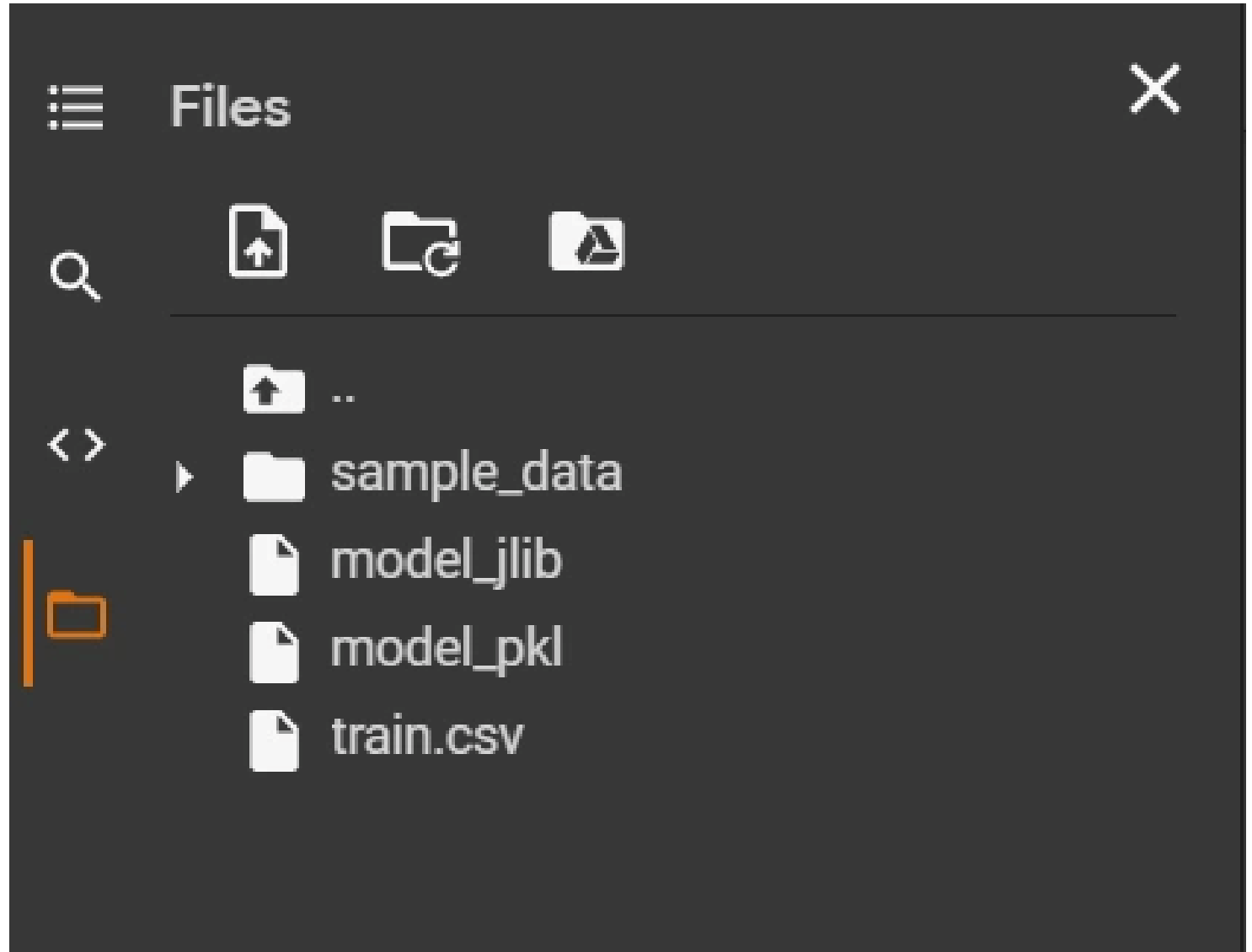
The Concept of Serialization



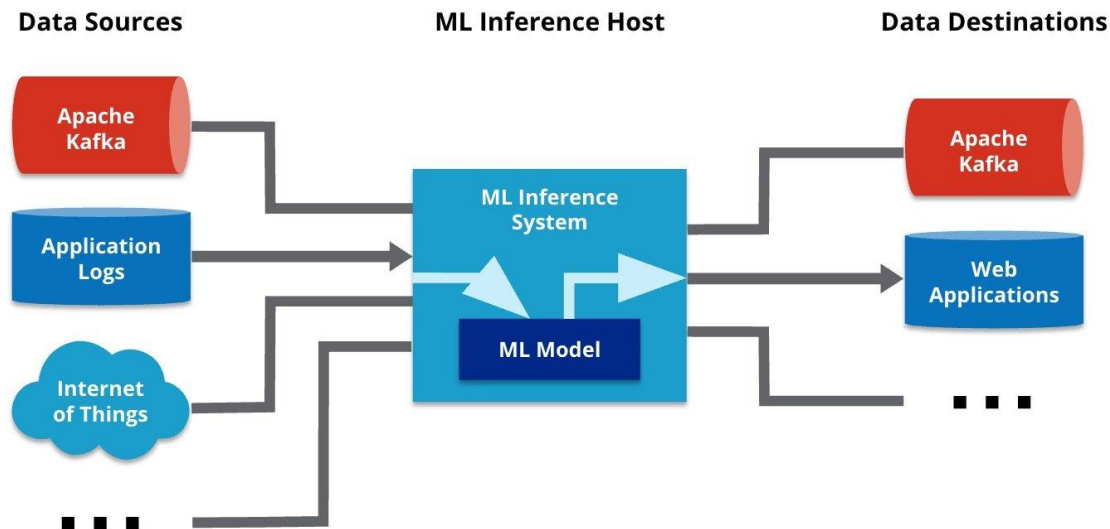
- **The Problem:** Training a model can take minutes, hours, or even days depending on the data size. You don't want to retrain the model every time you want to make one prediction.
- **The Solution:** Serialization.
- **Definition:** Converting the trained model object into a byte stream (a file) that can be stored on a hard drive and reloaded later.

Using Pickle & Joblib

- **Pickle:** The standard Python library for serializing objects.
- **Joblib:** A library optimized specifically for large NumPy arrays and Scikit-Learn models. It is usually faster and more efficient for ML models than standard Pickle.
- **Files:** These models are typically saved with the .pkl or .joblib extension.



The Inference Script



- **Inference:** The process of using a trained model to make a prediction on new, live data.
- **The Script:** In a production environment, you write a separate, lightweight script that:
 - Loads the saved model file.
 - Accepts input from a user (like a website form or a sensor).
 - Outputs the prediction instantly without any training code involved.